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Unit-4

Medicinal Chemistry - IInd

Drugs Acting on Endocrine System

• Syllabus →

Nomenclature, stereochemistry and metabolism of steroids

- ① Sex hormones → Testosterone, Nandrolone, Progesterone, Oestriol, Oestradiol, oestriol, Diethyl stilbestrol
- ② Drugs used for erectile dysfunction → Sildenafil, Tadalafil.
- ③ Oral contraceptives → Mifepristone, Norgestrel, Levonorgestrel.
- ④ Corticosteroids → Cortisone, Hydrocortisone, Prednisolone, Betamethasone, Dexamethasone
- ⑤ Thyroid and Antithyroid drugs → L-Thyroxine, L-Thyronine, Propylthiouracil, methimazole

• Important Questions →

- 1) Write a note on oral contraceptives.
- 2) Define and classify Antithyroid drugs and also write a note on their drug propylthiouracil and L-thyroxine.
- 3) Write a short note of any two:-
 - i) steroids
 - ii) Corticosteroids
 - iii) Sex hormones
 - iv) Drugs for erectile dysfunction.

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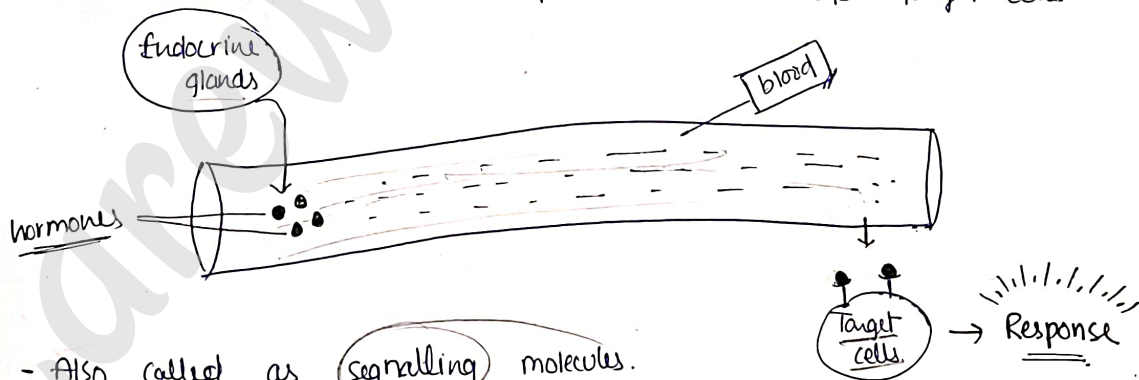
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Endocrine system →

- It is the collection of glands that produces hormones that regulate many physiological function in body such as growth, metabolism, sexual function etc--
- It also helps to maintain internal homeostasis by the use of hormones.
- These are ductless glands that directly secrete their hormone product into the blood.

Hormone → It is a substance of intense biological activity that is produced by specified cells (endocrine glands) in the body and it is transported through circulation to act on its target cells.



- Also called as signalling molecules.

Properties →

- they act in very low concentration.
- produce their effect on target cells
- Rapidly destroyed after their function over.
- It may be peptides, steroids; amines etc--

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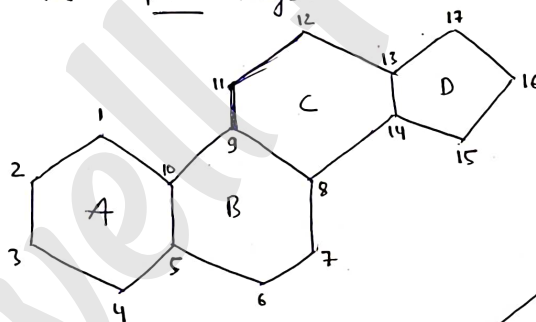
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- ① Define steroids and their nomenclature, stereochemistry and metabolism.

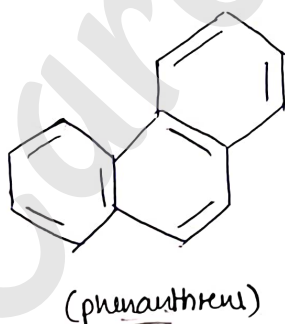
[OR]

Nomenclature, Stereochemistry and Metabolism of steroids.

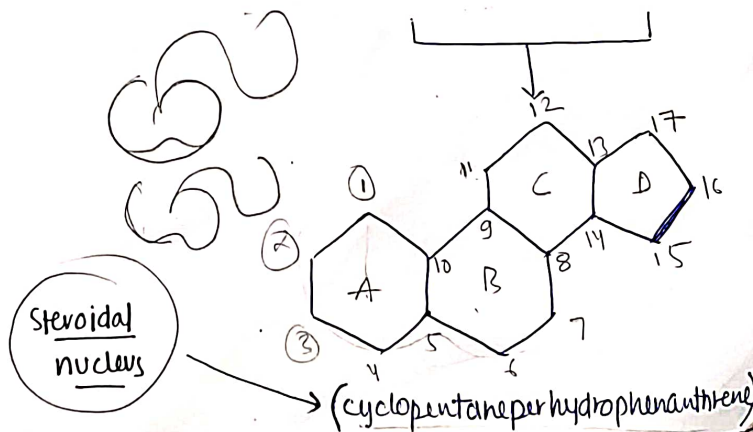
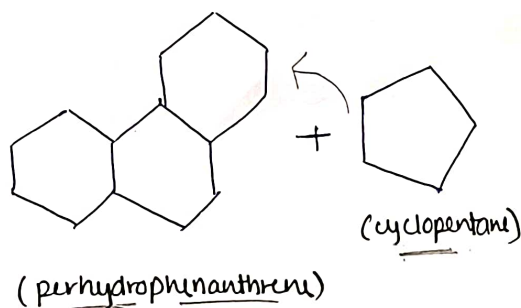
- steroids → These are the class of naturally occurring organic compounds produced naturally in the body.
- they are used for many physiological function in body, such as growth and development.
 - They have a steroidal ring which contain 17 carbon in four rings.



• Nomenclature →



hydrogenation →



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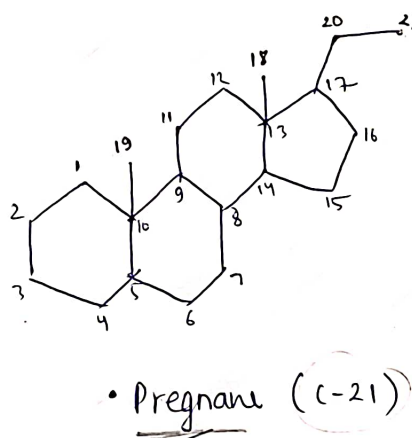
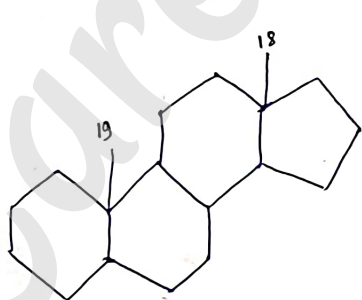
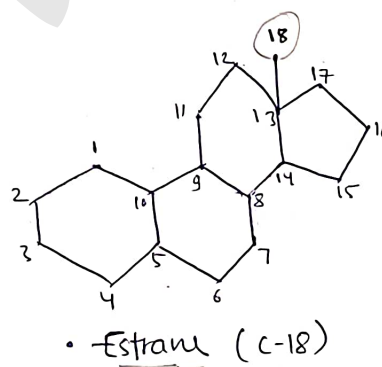
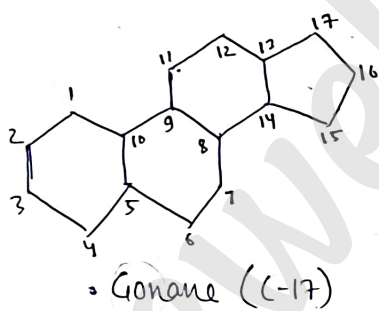
- The basic skeleton contains 17 Carbons atoms and the numbering starts with ring A

- Ring A
 - Ring B
 - Ring C
- } Cyclohexane ring and Ring D → Cyclopentane ring

or complete

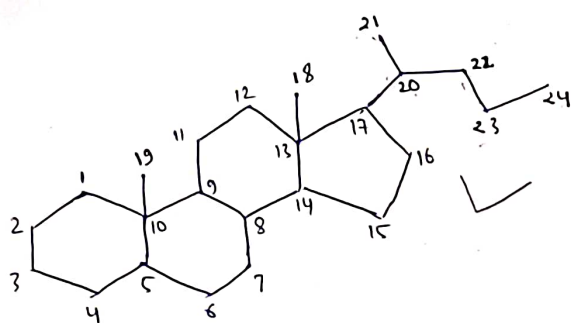
Ring A, B, C + Ring D
 perhydrophenanthrene Cyclopentane

Cyclopentanoperhydrophenanthrene

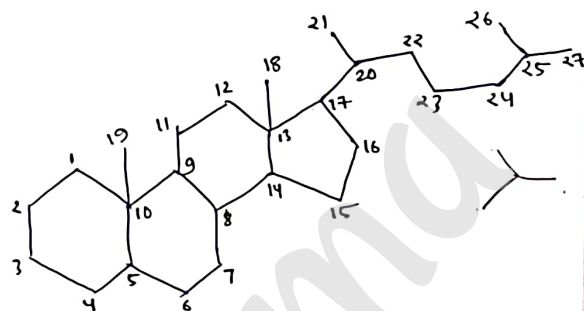


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cholan (C-24)

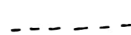


cholesterol (C-27)

- Solid line indicates groups above the plane of the nucleus (β -configuration) and dotted line denote groups below the plane (α -configuration).

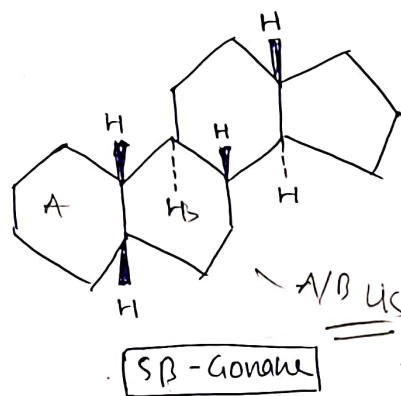
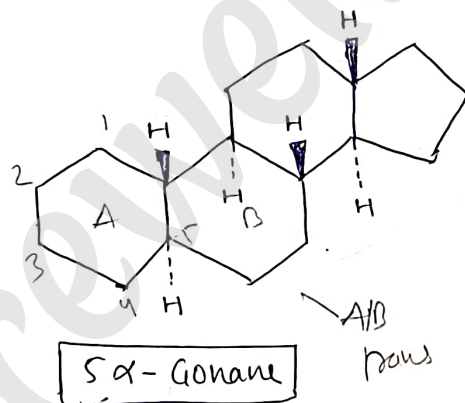


→ solid line → β -configuration



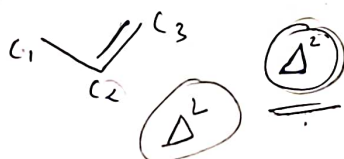
→ dotted line → α -configuration

(eg)



- The symbol Δ (delta) is used to indicate C=C bond in steroids.

(eg)

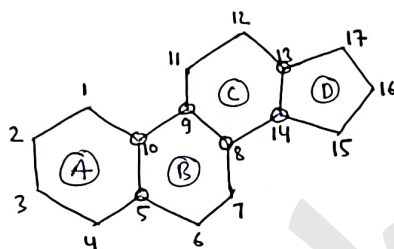


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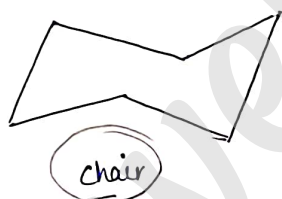
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• stereochemistry →

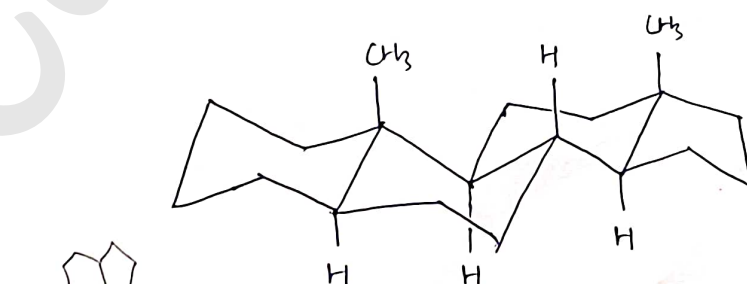
there are six asymmetric carbon atoms 5, 8, 9, 10, 13, 14 in the nucleus, therefore 64 optically active forms are possible



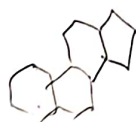
- They are mostly exists in two conformation such as chair form and boat form



→ chair form is more stable than boat form due to less angle strength, therefore all cyclohexane ring in steroid nucleus exist in the chair form.



— chair form
(mostly steroids)



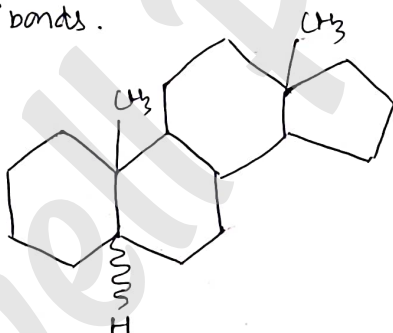
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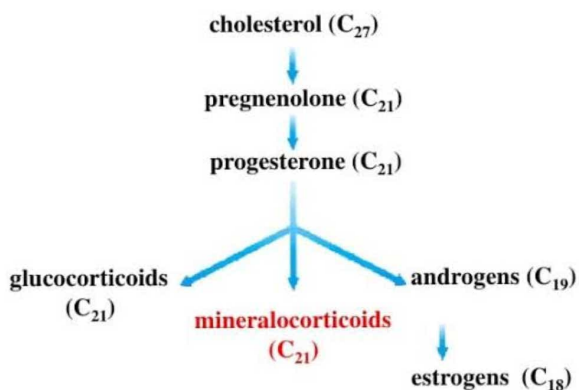
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- Also, Hydrogen or functional group on the β -side of the molecules are denoted by solid lines and the α -side is denoted by dotted line.
- S- α -steroids $\rightarrow$ A/B trans
- S- β -steroids $\rightarrow$ A/B cis
- If A/B fusion cis and trans both position possible or position or unknown, it is indicated by waving lines/bonds.



- Metabolism $\rightarrow$ primarily oxidized by CYP P450 oxidase enzymes such as CYP3A4.



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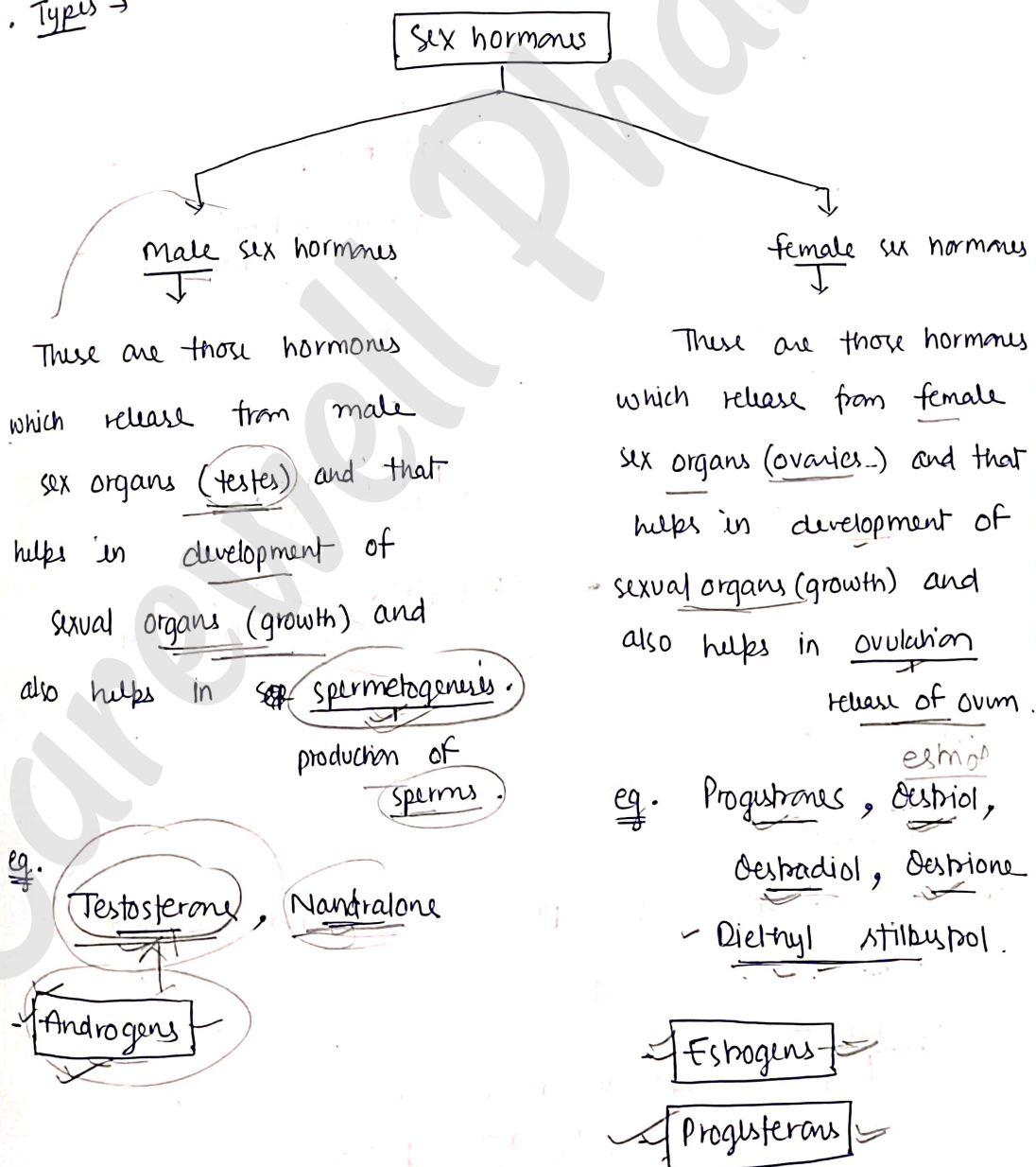
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② write a short note on Sex hormones.

These are those hormones which produce and released in human body and they play an important role in sexual development & reproduction.
 other names → sex steroids, Gonadocorticoids, Gonadal steroids.

Types →



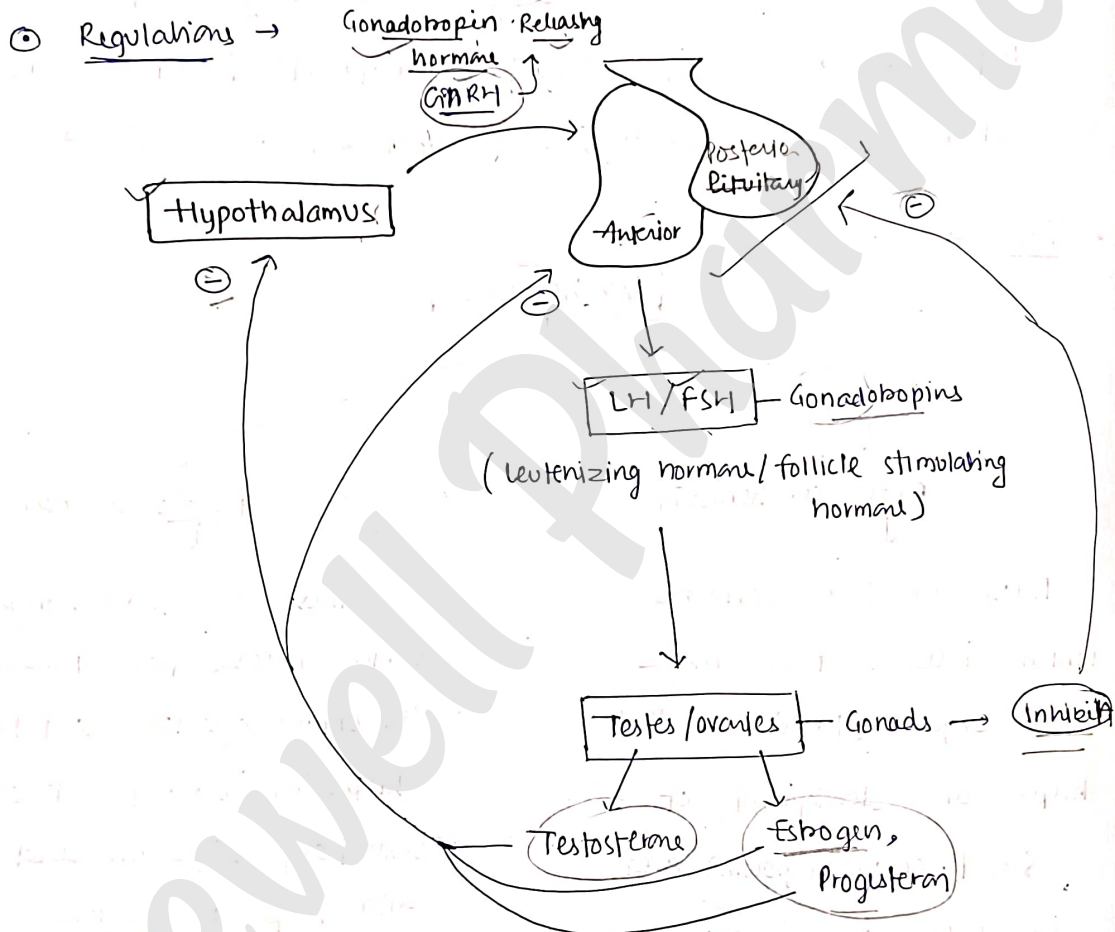
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- All these hormones are synthesized from cholesterol mainly in the gonads and adrenal cortex.



DRUGS →

- ① Testosterone → catabolic to anabolic - Bone ↑, muscle ↑
 • It is the primary sex hormone and anabolic steroids.

functions →

- Development of male reproductive tissues such as testes and prostate
- promotes sexual characteristics such as increased muscles and bone mass, body hair etc..

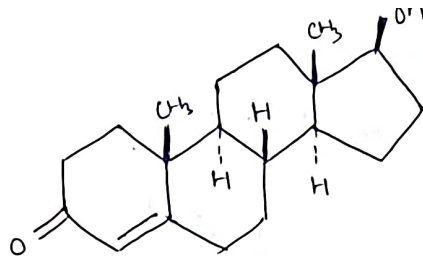
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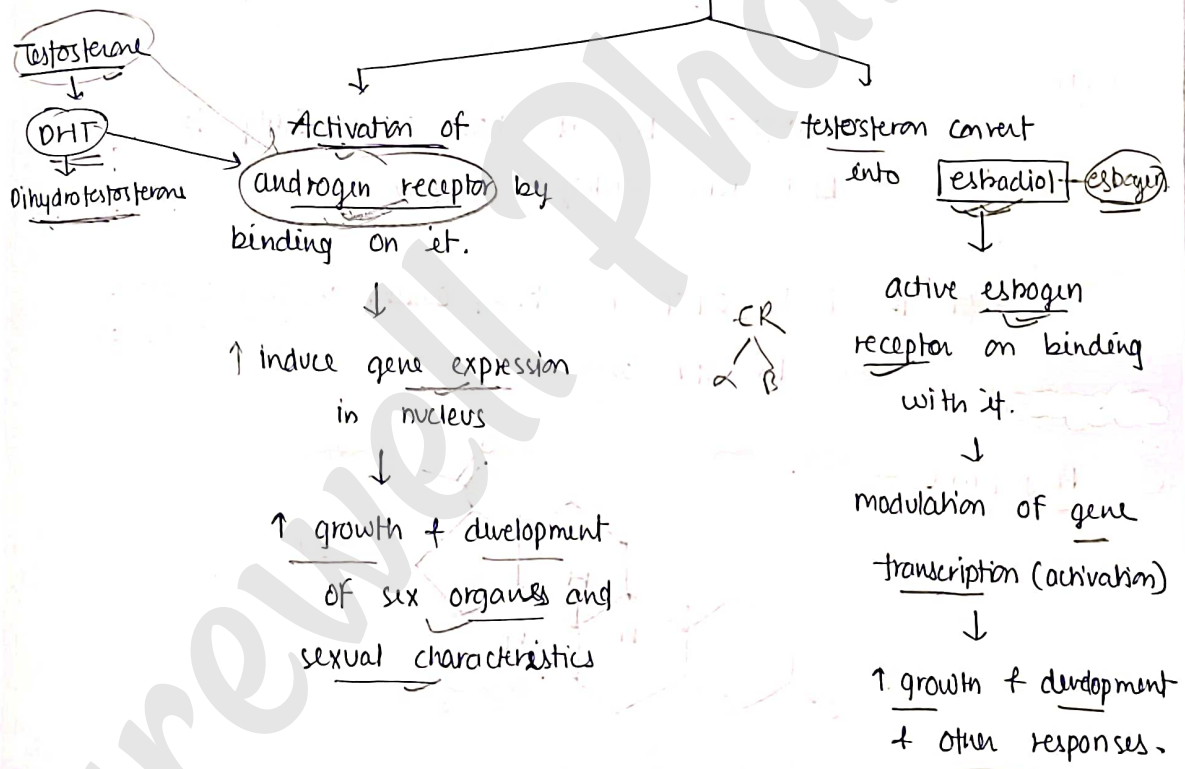
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• structure →



• IUPAC Name → 17β -Hydroxyandrost-4-en-3-one

• MOA → It gives their action by two main mechanisms



• Uses →

- It is used primarily to treat symptoms of sexual dysfunctions.
- It is used to increase bone mass and increased sense of well-being.

• Advs →

- Acne, swelling and breast enlargement in men.
- Liver toxicity, heart disease and behavioral changes.

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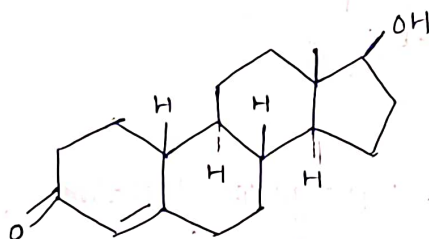
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② Nandrolone → (19-nortestosterone)

- almost same as testosterone



IUPAC → 19-norandrost-4-en-17 β -ol-3-one

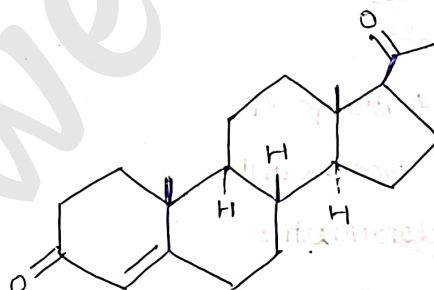
MOA →

same as testosterone but it has great ability to bind with androgen receptor.

Uses →

used to treat osteoporosis, breast cancer, wasting syndrome (cachexia) and also for anemia.

③ Progesterone →



MOA → (function) →

- It binds with both receptor (progesterone & also estrogen).
- It converts the endometrium to its secretory stage to prepare the uterus for implantation.
- It also thick the cervical mucus in cervix.

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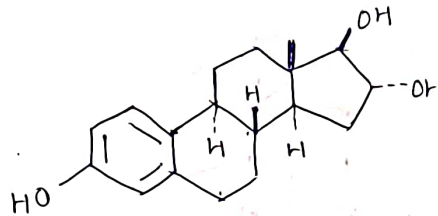
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Uses → used in oral contraceptives pills.

Estrogen → Estane (E_1)
Estadiol (E_2)
Estiol (E_3)

④ Oestriol →
 estriol



MOA → Oestriol → bind with → Estrogen Receptor [ER] $\begin{cases} ER-\alpha \\ ER-\beta \end{cases}$
 (Nuclear steroid hormone receptor)

Growth & development and other responses.

modulation of gene transcription in Nucleus

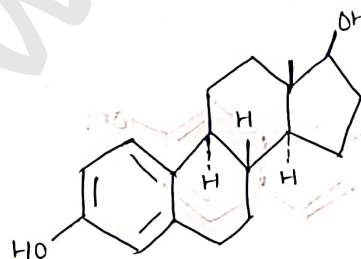
Activates

Uses →

Used in hormone therapy for menopausal symptoms.

for less potent than oestradiol and weak estrogen

⑤ Oestradiol →



MOA → same as oestriol, but more potent than oestriol.

Uses →

- It is used to treat menopause symptoms such as hot flashes and vaginal changes
- used to prevent osteoporosis (bone loss) in menopausal women.
- also used to treat low estrogen levels in women with

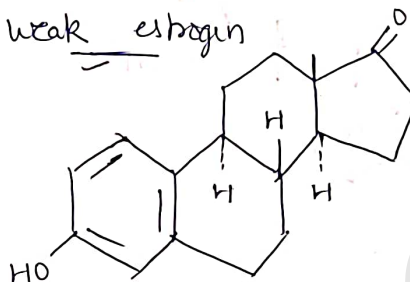
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ovarian failure

⑥ Oestrone → (oestron).

- weak estrogen



MOA → same (but it is ^{for} less potent than estradiol)

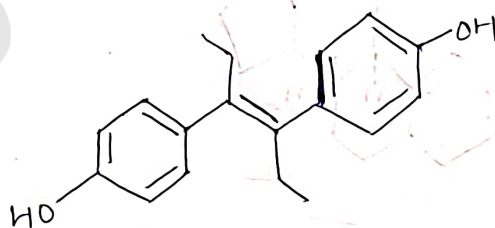
uses → same for menopausal therapy (but not mostly used)

- used as an injected estrogen

⑦ Diethyl stilboestrol →

also known as diethyl stilboestrol.

It is synthetic, nonsteroidal estrogen medication.



MOA →

It inhibits the hypothalamic-pituitary-gonadal axis, & decreasing the secretion of GnRH.

uses → mostly used in the treatment of prostate cancer and less commonly in breast cancer.

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③ Write a note on Drugs for erectile dysfunction.

- Erectile dysfunction (ED) is a male disorder characterized by inability of a male to achieve or maintain an erection of penis long enough to engage in sexual intercourse.
- It may affect a relationship negatively and break a person's self confidence when the condition tends to be ongoing.
- Patients who suffer from erectile dysfunction either totally fail to achieve an erection of penis, or are not able to consistently maintain it.
- Causes →

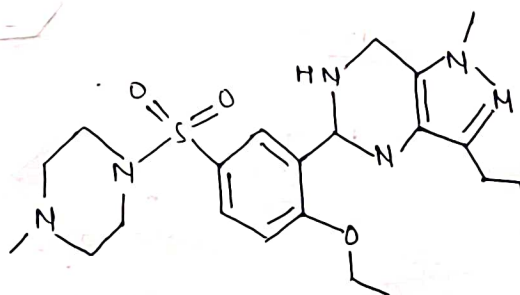
- When blood doesn't flow well to the penis.
- Underlying psychological and health conditions.
- Trauma and unhealthy life style

stress and anxiety
lack of sleep
smoking, drinking
↑BP, diabetes etc
old age.

- Drugs →
- Sildenafil
 - Tadalafil

i) Sildenafil →

- It is sold under brand name Viagra.
- most effective when taken on an empty stomach one hour before sex.



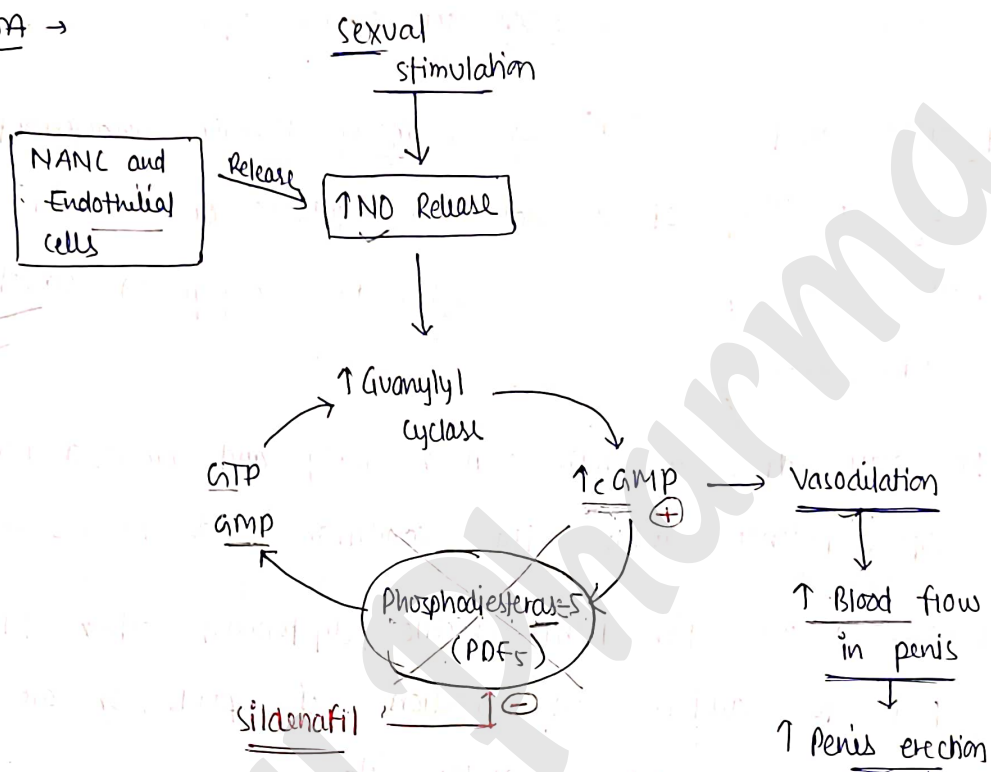
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MOA →



- It act by blocking phosphodiesterase-5 (PDE5), an enzyme that promotes breakdown of cGMP, which regulate blood flow in the penis.

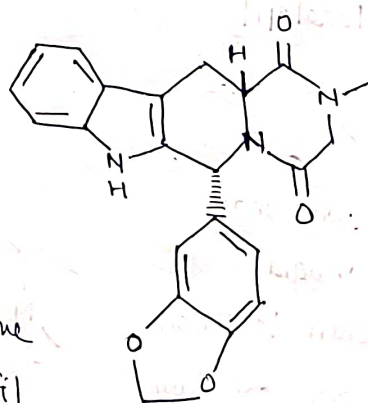
Uses →

- used to treat male sexual function problems (impotence or erectile dysfunction etc).

ii) Tadalafil →
(Cialis)

MOA &
Uses

almost same
as sildenafil.



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④ Write a details note on Oral contraceptives

→ Oral contraceptives →

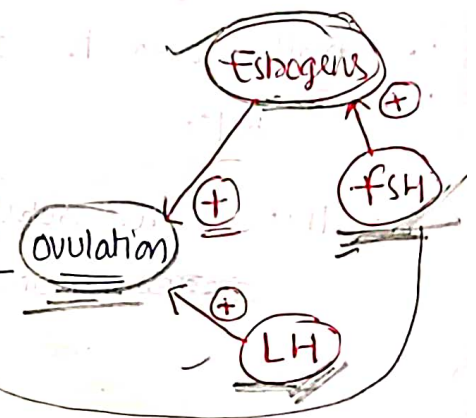
Also known as "Birth control Pills".

• These are those drugs which are used to prevent pregnancy. prescription medications

• These medicines contain hormones like estrogen and progestin (synthetic but similar as progesterone)

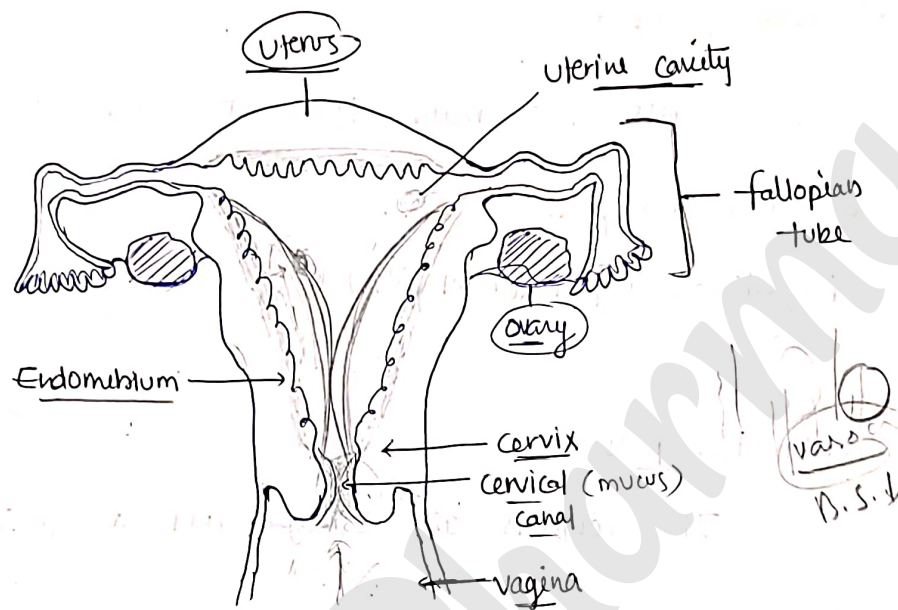
• Role of these hormone
in pregnancy [normal mechanism] →

— The ovaries produce the egg cells called ovum (ova) then it transported into fallopian tube.
release



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- Now this egg fertilized by sperm in fallopian tube. (*)
- Now, this egg moves to uterus, where the uterine lining has thickened and prepare to receive a fertilized egg. (+)



- Rising level of progesterone that cause glands in lining to release fluids that feeds the egg. (fertilized egg). endometrium
- It also causes the thinned out mucus in the cervix to become thick again which prevent sperm from passing through.

If an egg hasn't been fertilized the level of both estrogen and progesterone begins to fall which cause menstruation.

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- Oral contraceptives →

there are two main types of oral contraceptives —

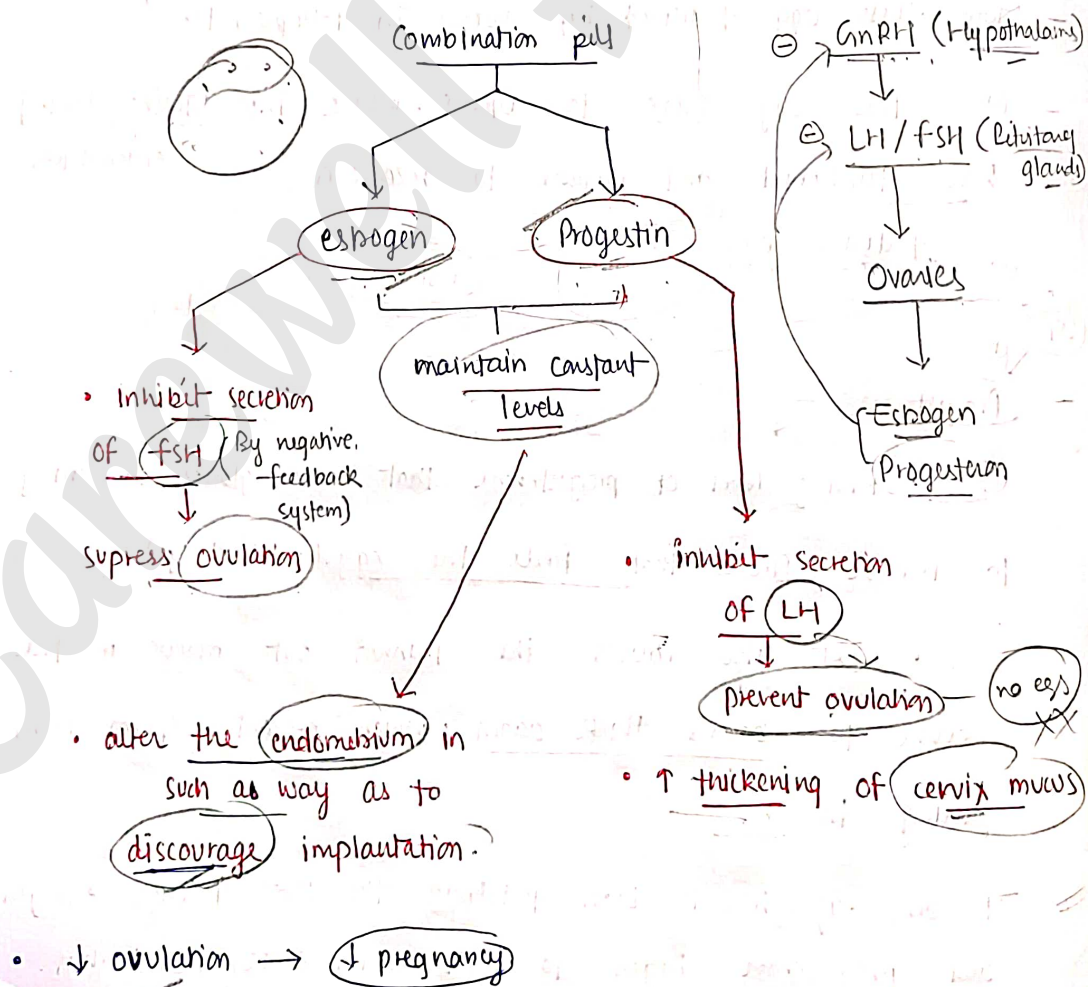
- i) Combination pills
- ii) Progestin-only pill (mini pill)

- i) Combination pills →

It contains an estrogen and a progestin

Synthetic version of progesterone.

MoA →



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• this is most popular and most efficacious method.

eg) Ethinylestradiol & Norgestrel etc —

Adve → weigh gain, nausea, vomiting etc —

10) Course → Day wise → < 1 active pills that contain hormones.

11) Minipill (progestin only pill) →

They contain only progestin.

MOR → prevent ovulation & thickening of cervical mucus
 prevent pregnancy

— this efficacy is lower than combination pills.

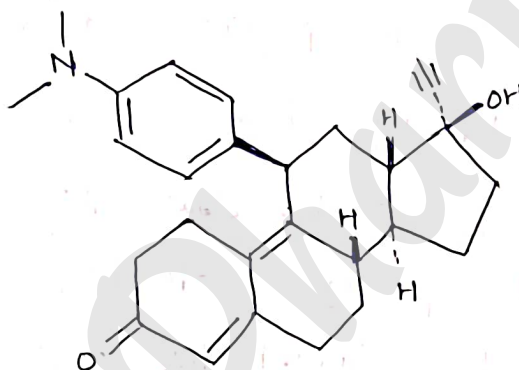
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Drugs → Mifepristone , Norgestrel , Tadalafil

i) Mifepristone →

It is a synthetic steroids and also known as RU-486.



MOA → Combination pills

- It blocks the action of progesterone which alter the endometrium (the uterine lining), induce bleeding, and causes the uterine lining to shed.

Uses →

It is used in combination with misoprostol (cytotox) to end an early pregnancy.

ADR →

Nausea, vomiting, bleeding, weakness, dizziness etc -

ii) Norgestrel →

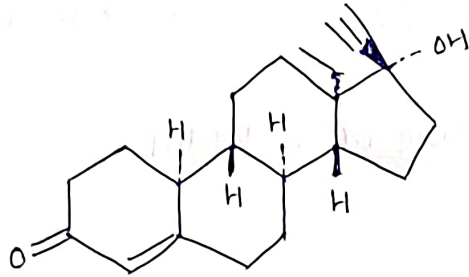
It is a form of progestin hormone that prevents pregnancy.

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MOA →

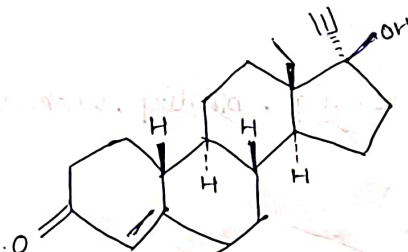
- It binds to the progesterone and estrogen receptor within the female reproductive tract and other sites.
- After binding it slows down the release of GnRH and LH by negative feedback system
- which further inhibits the ovulation & pregnancy.
- It also increases the thickness of cervical mucus in cervix which inhibits the entry of sperm.

Uses →

- used in combination with ethinylestradiol/quinestrol in combined pill.
- also used alone in progesterin-only birth pills
- in combination with estradiol in menopausal hormone therapy

iii) Levonorgestrel →
(LNG)

MOA → same as Norgestrel



Uses →

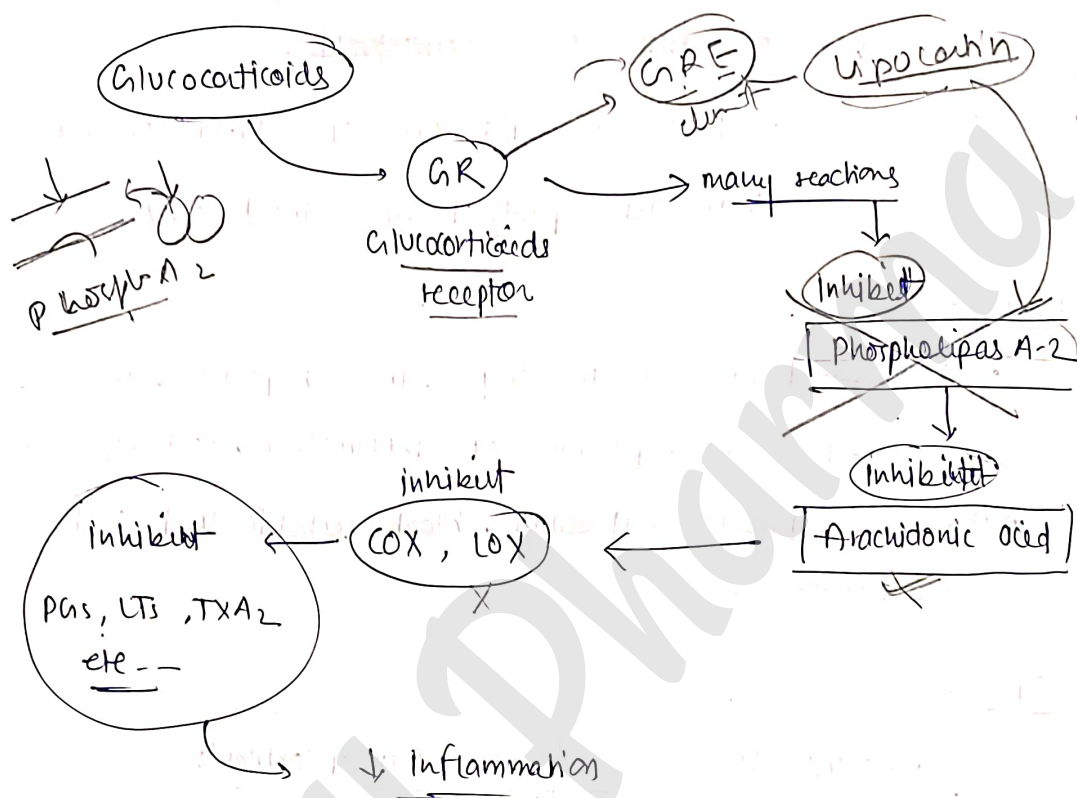
- sold under brand name Plan B. it is useful within 120 hrs as emergency birth control.
- Also used in combination with estrogen.

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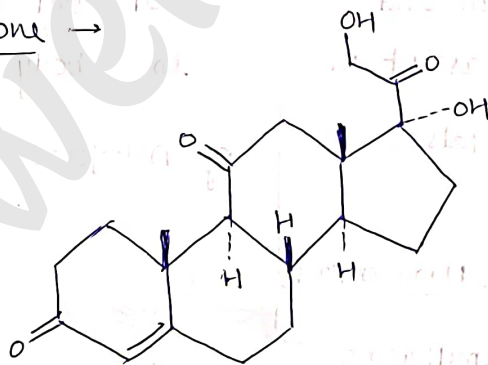
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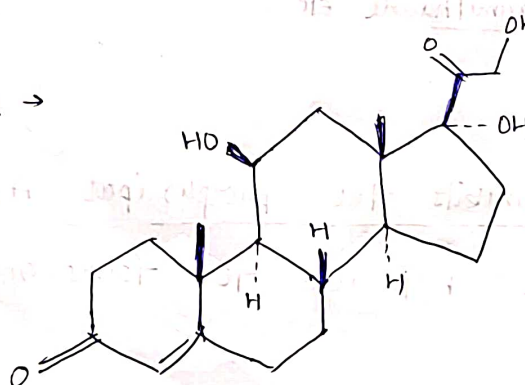
①

Cortisone →



②

Hydrocortisone →



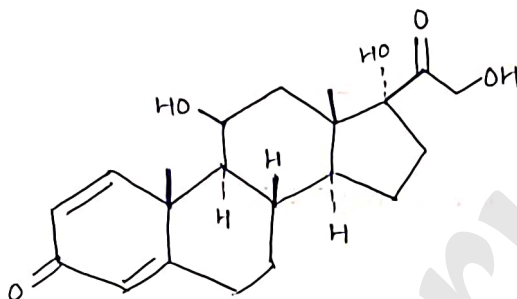
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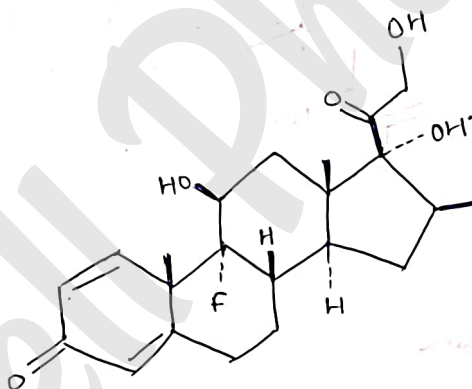
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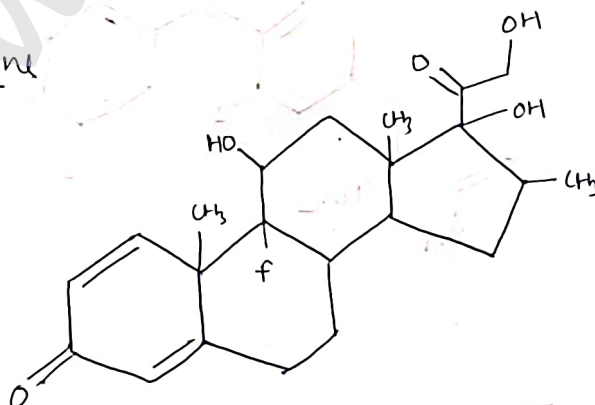
③ Prednisolone



④ Bitamethasone



⑤ Dexamethasone



uses → used as an anti-inflammatory agents.

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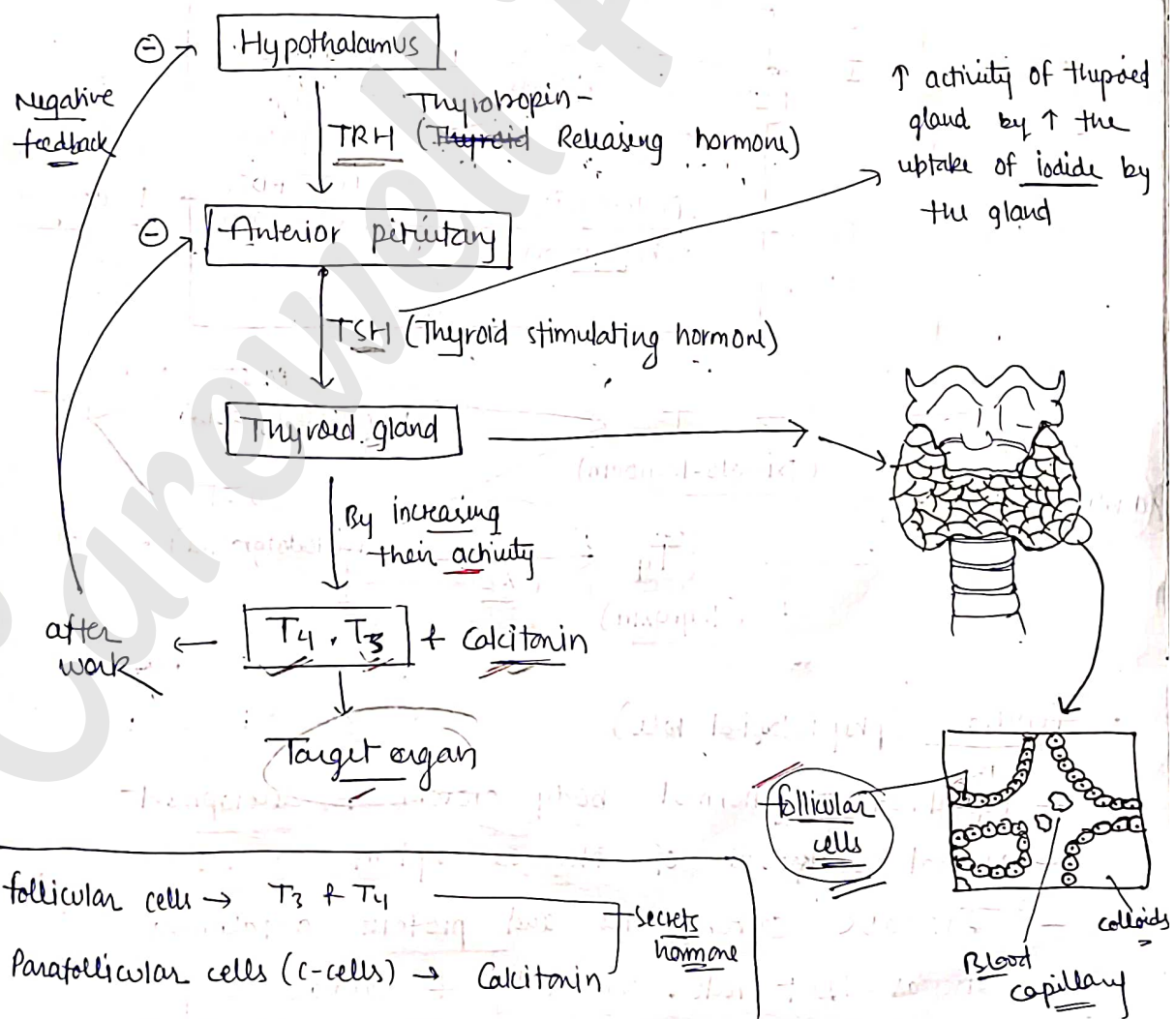
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⑥ Discuss in details about thyroid and Antithyroid drugs.

- Thyroid hormones → Those hormones which are secreted from thyroid glands and are responsible for body's growth and work.

Thyroid gland → A butterfly shaped gland which present on neck in front of trachea



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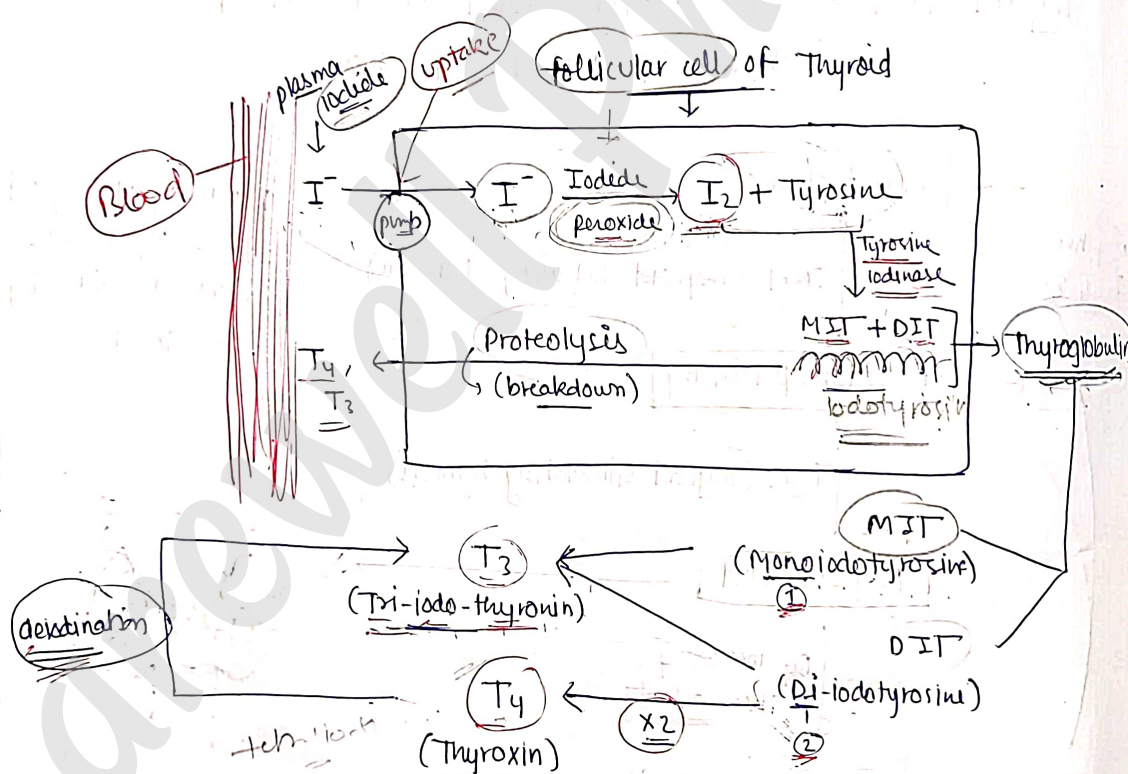
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- Synthesis and Release of thyroid hormone.

- Uptake of iodide
- Oxidation of iodide to free iodine & iodination of tyrosine.
- Coupling of iodotyrosines.
- Proteolysis of thyroglobulin and release of T₄ + T₃
- Conversion of T₄ to T₃ by deiodination.



- Functions (physiological role)

- Essential for normal body growth & development
- Normal development of Nervous system
- Increase carbohydrates and protein metabolism
- Increase heart rate, contractility & cardiac output.

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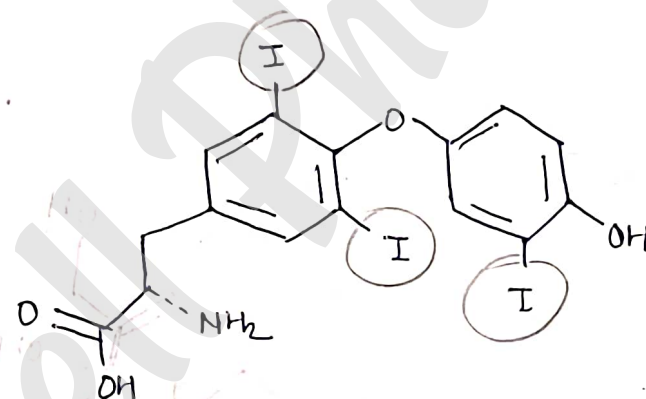
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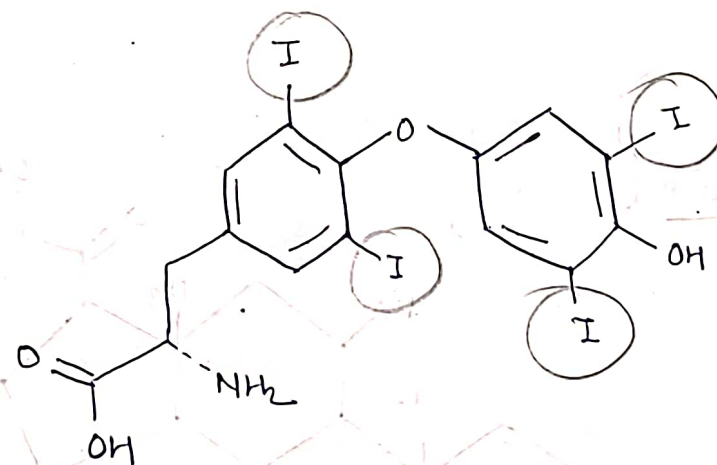
Thyroids & Antithyroids

- Structures →

(T₃) → Tri-iodo-thyronine



(T₄) → Thyroxine



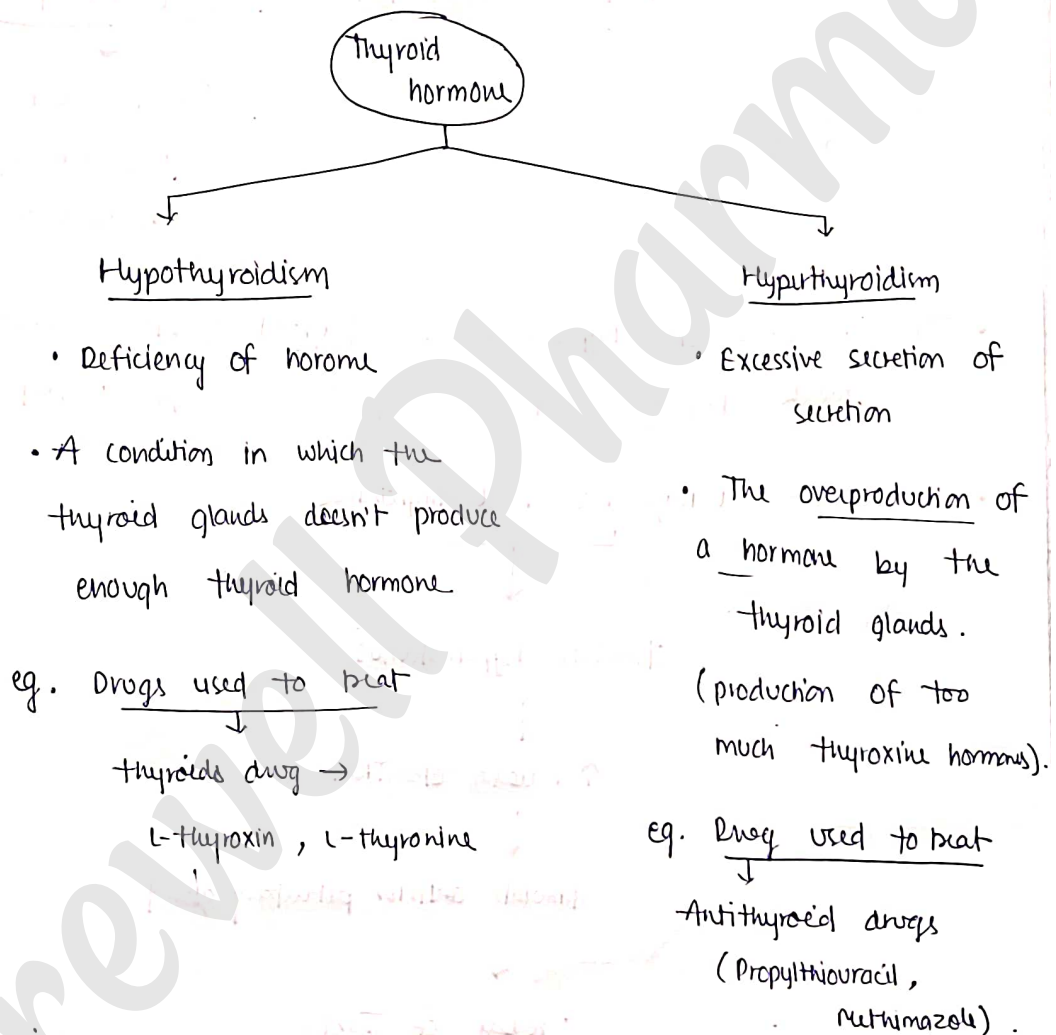
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- It also have hypolipidemic effects and it also raise basal metabolic rate (BMR).



① Thyroid drugs →

These are those drugs which ^{are} used in the treatment of hypothyroidism in which they increase the activity of thyroid gland.

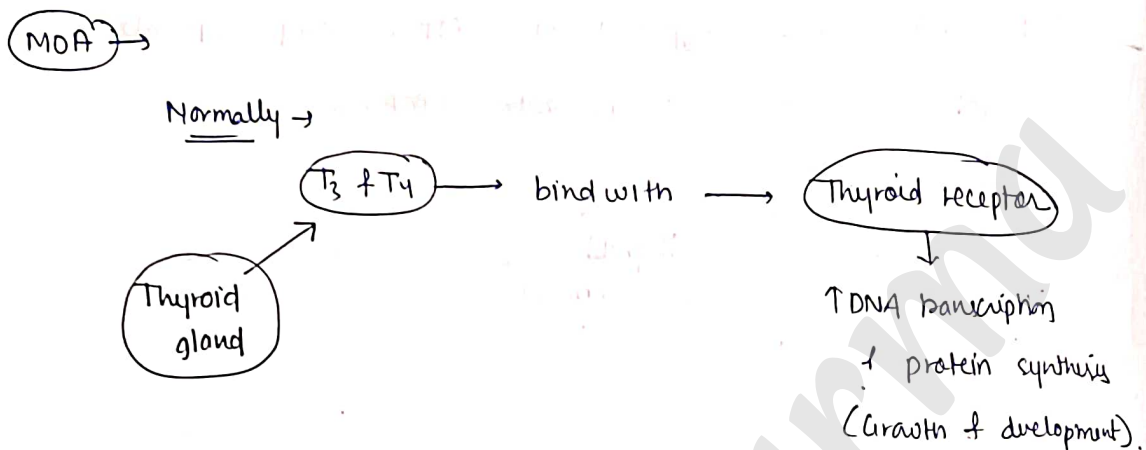
Drugs → L-thyroxine (levothyroxin) , L-thyronine (liothyronine)

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- In the case of hypothyroidism, thyroid gland doesn't work properly and not produce enough thyroid hormone.

- Introduce L-thyroxine & L-thyronine

Stimulate hypothalamus

↑ release of TRH

Stimulate anterior pituitary gland

↑ release of TSH

Thyroid gland activity

increase

↑ production of
T₃ & T₄

Response

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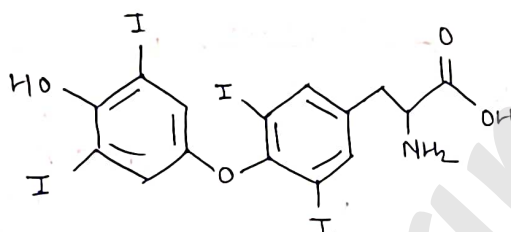
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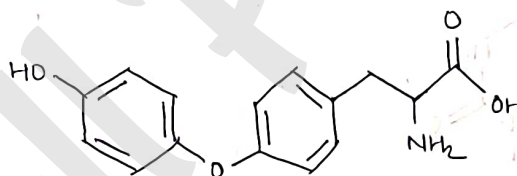
i) L-thyroxine →

Also known as L-thyroxine. It is a synthetic form of thyroxine (T_4).



ii) L-thyronine →

Synthetic form of T_3 .



both are
(US) → used to treat hypothyroidism.

② Antithyroid drugs →

- These are those drugs which inhibit the production & release of thyroid hormone by controlling the overproduction of thyroid hormone.
- used to treat hyperthyroidism.

(MOA) →

i) Propylthiouracil ii) methimazole

Both are belongs to the class of (inhibit hormone synthesis) and also thioamides.

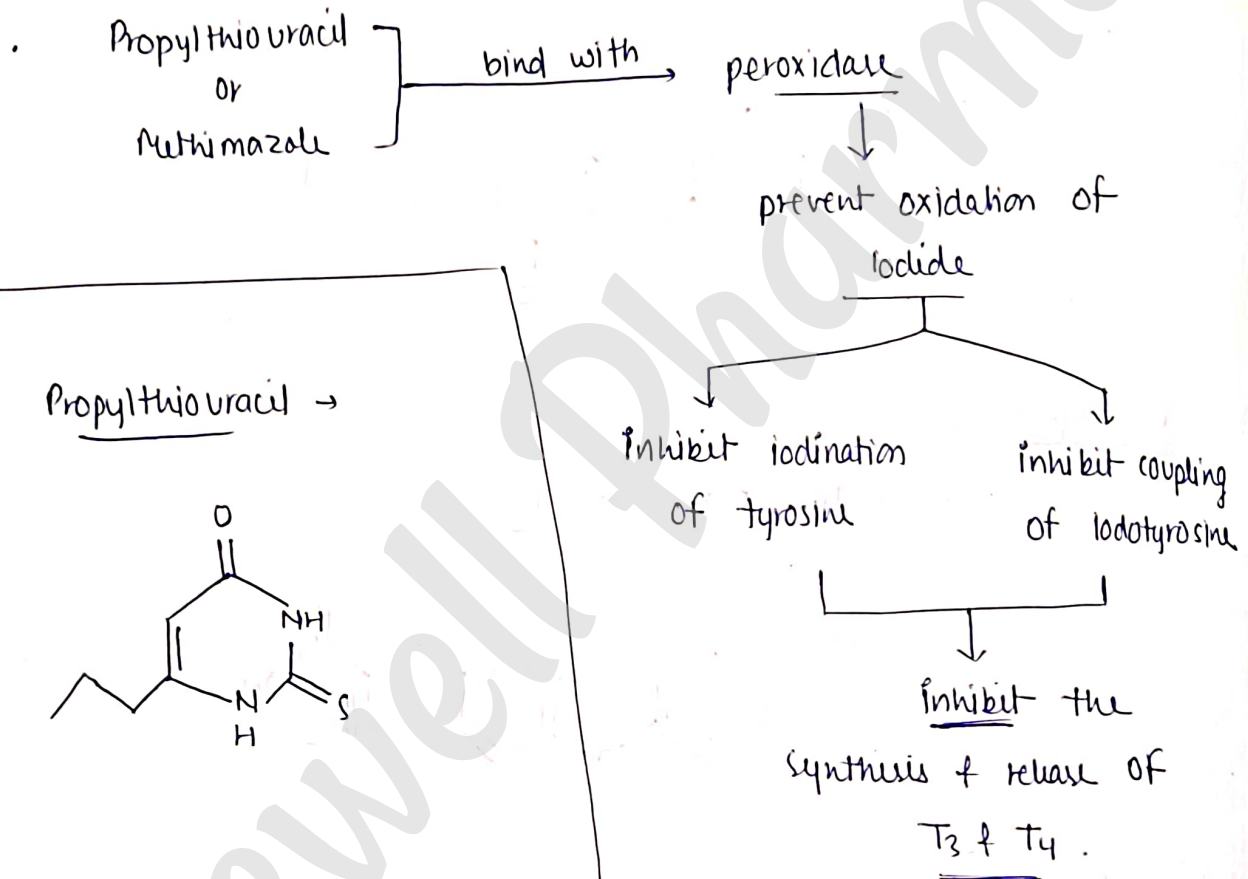
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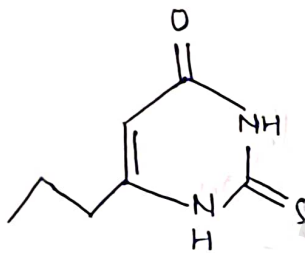
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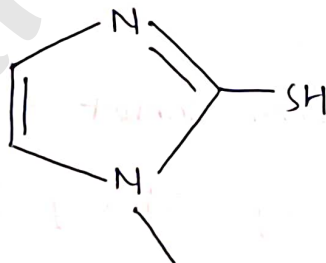
- they give their action by inhibiting the synthesis of thyroid hormones.



i) Propylthiouracil →



ii) Methimazole →



(Uses) →

Used in the treatment
of hyperthyroidism.

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